



Install Required Libraries

```
In [2]: import nltk
nltk.download('stopwords')
nltk.download('punkt')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
```

```
Out[2]: True
```

Import Libraries

```
In [3]: import pandas as pd
import re
import matplotlib.pyplot as plt

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

from sklearn.feature_extraction.text import TfidfVectorizer
from wordcloud import WordCloud
```

Loading dataset

```
In [4]: df = pd.read_csv("/content/Tweets.csv")
df
```

Out[4]:

	tweet_id	airline_sentiment	airline_sentiment_confidence	neg
0	570306133677760513	neutral	1.0000	
1	570301130888122368	positive	0.3486	
2	570301083672813571	neutral	0.6837	
3	570301031407624196	negative	1.0000	
4	570300817074462722	negative	1.0000	
...	...	...	...	...
14635	569587686496825344	positive	0.3487	
14636	569587371693355008	negative	1.0000	
14637	569587242672398336	neutral	1.0000	
14638	569587188687634433	negative	1.0000	
14639	569587140490866689	neutral	0.6771	

14640 rows × 5 columns

Filter Negative Sentiment Tweets

```
In [5]: negative_df = df[df['airline_sentiment'] == 'negative']
negative_df = negative_df[['text']]
negative_df.head()
```

```
Out[5]:
```

	text
3	@VirginAmerica it's really aggressive to blast...
4	@VirginAmerica and it's a really big bad thing...
5	@VirginAmerica seriously would pay \$30 a fligh...
15	@VirginAmerica SFO-PDX schedule is still MIA.
17	@VirginAmerica I flew from NYC to SFO last we...

Text Preprocessing

```
In [8]: stop_words = set(stopwords.words('english'))
nltk.download('punkt_tab')

def clean_text(text):
    text = re.sub(r"http\S+", "", text)      # remove URLs
    text = re.sub(r"@w+", "", text)         # remove mentions
    text = re.sub(r"#w+", "", text)         # remove hashtags
    text = re.sub(r"[^a-zA-Z\s]", "", text)  # remove special chars
    text = text.lower()

    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]

    return " ".join(tokens)

negative_df['cleaned_text'] = negative_df['text'].apply(clean_text)
negative_df.head()
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
```

```
Out[8]:
```

	text	cleaned_text
3	@VirginAmerica it's really aggressive to blast...	really aggressive blast obnoxious entertainmen...
4	@VirginAmerica and it's a really big bad thing...	really big bad thing
5	@VirginAmerica seriously would pay \$30 a fligh...	seriously would pay flight seats didnt playing...
15	@VirginAmerica SFO-PDX schedule is still MIA.	sfopdx schedule still mia
17	@VirginAmerica I flew from NYC to SFO last we...	flew nyc sfo last week couldnt fully sit seat ...

Compute TF-IDF

```
In [9]: tfidf = TfidfVectorizer(max_features=20)
tfidf_matrix = tfidf.fit_transform(negative_df['cleaned_text'])

tfidf_df = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=tfidf.get_feature_names_out()
)

tfidf_df.head()
```

```
Out[9]:
```

	amp	call	cancelled	cant	customer	delayed	flight	flightled	get	help	hc
0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(
2	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	(
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	(

Identify Top TF-IDF Terms

```
In [10]: tfidf_scores = tfidf_df.sum().sort_values(ascending=False)

tfidf_scores
```

Out[10]:

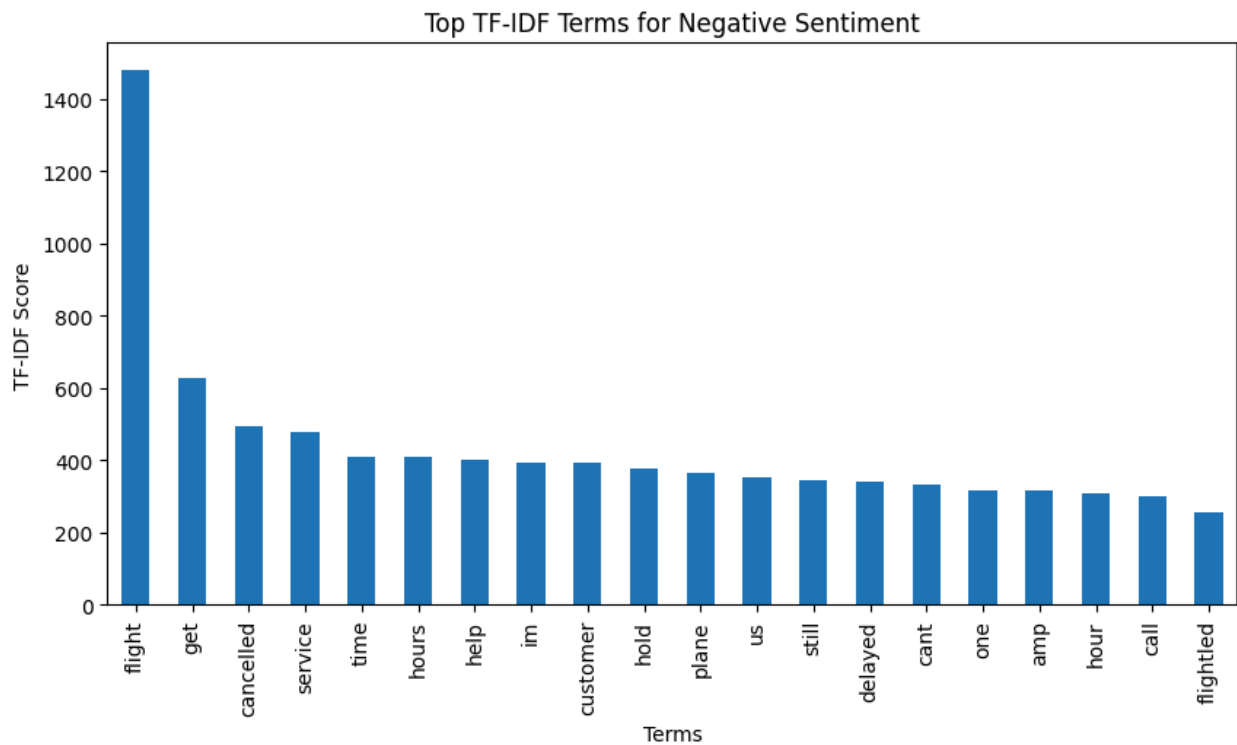
0

flight	1481.000000
get	628.834187
cancelled	495.030891
service	480.297397
time	410.137949
hours	408.907301
help	400.960459
im	393.561573
customer	391.901305
hold	375.947292
plane	365.325339
us	354.717077
still	346.068281
delayed	340.920914
cant	333.578100
one	317.401553
amp	316.283743
hour	308.308436
call	299.907081
flightled	255.408502

dtype: float64

Bar Chart Visualization

```
In [11]: plt.figure(figsize=(10,5))
tfidf_scores.plot(kind='bar')
plt.title("Top TF-IDF Terms for Negative Sentiment")
plt.xlabel("Terms")
plt.ylabel("TF-IDF Score")
plt.show()
```



Word Cloud Visualization

```
In [12]: wordcloud = WordCloud(  
    width=800,  
    height=400,  
    background_color='white'  
)  
.generate_from_frequencies(tfidf_scores)  
  
plt.figure(figsize=(10,5))  
plt.imshow(wordcloud, interpolation='bilinear')  
plt.axis('off')  
plt.show()
```



Discussion Session

In []: `## Discussion`

```
# In this experiment, TF-IDF was applied to negative airline tweets to
# identify the most important words contributing to negative sentiment.
# The analysis highlights frequent complaint-related terms such as delays,
# service, and cancellations. TF-IDF helps in emphasizing sentiment-specific
# vocabulary by reducing the impact of commonly occurring words.
```